

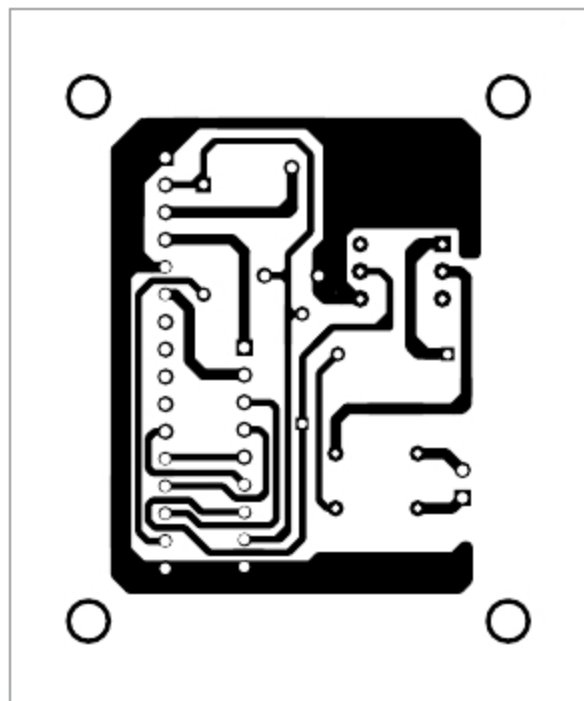


Fig. 4: Actual-size PCB layout of digital frequency meter

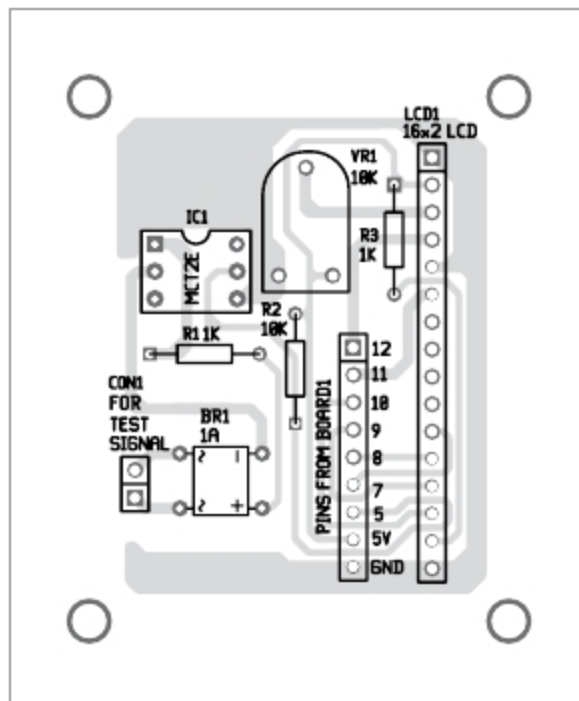


Fig. 5: Components layout for the PCB

used as pulse input pin, and digital I/O pins 7, 8, 9, 10, 11 and 12 of Arduino Uno board are used to interface with the LCD.

Bridge rectifier. The bridge rectifier (BR1) is used to rectify the alternating input signal to pulsating DC.

Optocoupler. The optocoupler is used to produce spikes from the pulsating DC waveform.

16x2 LCD. The 16x2 LCD is used to display the frequency. It consists of 16 columns and two rows for displaying data. The LCD used here is a backlit one. It is interfaced with

Arduino in 4-bit mode. LCD pins RS, EN, D4, D5, D6 and D7 are connected to digital I/O pins 12, 11, 7, 8, 9 and 10 of Arduino, respectively.

Software

Arduino IDE is used for programming Arduino Uno. Open the source code/sketch (Freq_meter_1.ino), and select the COM port and board from Tools menu in Arduino IDE. After selecting the correct COM port and board, upload the source code to the board.

Construction and testing

An actual-size PCB layout of the digital frequency meter is shown in Fig. 4 and its components layout in Fig. 5. Assemble the components on the PCB, as per the circuit diagram. After this, connect the USB cable to Arduino and give the test signal at CON1 and check the frequency on LCD1 and the serial monitor of Arduino IDE. **EFY**

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Uno board
- IC1 - MCT2E optocoupler
- BR1 - 1A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3 - 1-kilo-ohm
- R2 - 10-kilo-ohm
- VR1 - 10-kilo-ohm preset

Miscellaneous:

- CON1 - 2-pin connector
- LCD1 - 16x2 alphanumeric LCD
- Function generator for signal input

EFY Note

The source code of this project is available for free download at source.efymag.com

Shibendu Mahata is M.Tech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. He is interested in designing MCU-based real-time embedded signal processing and process control systems



Saikat Patra is passionate about electronics and MCU-based embedded system applications



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